Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Classify Sentinel-1 SAR as active or passive and explain the distinction.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Active: the radar transmits a signal and measures its return. (2 points).

2. (2 points) Which kind of resolution describes the ground area represented by a pixel?

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answer: Spatial resolution. (2 points).

3. (2 points) Does a shorter orbital period necessarily give an equally short revisit time for a specified ground point? Explain.

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answer: No. Revisit depends on orbit geometry, swath and pointing as well as orbital period. (2 points).

4. (2 points) Using c=3.0e8 m/s, find the frequency of light with wavelength 6.0e-7 m.

answer: 500000000000000 Hz (2 points).

Numerical target 5e+14; absolute tolerance 1e+12. Accepted units: , Hz.

5. (2 points) Use Wien's law with b=2.90e-3 m K to find peak wavelength at 290 K.

answer: 0.00001 m (2 points).

Numerical target 1e-05; absolute tolerance 2e-08. Accepted units: , m.

6. (2 points) A radar echo takes 2.0e-5 s for its round trip. With c=3.0e8 m/s, calculate range.

answer: 3000 m (2 points).

Numerical target 3000; absolute tolerance 6. Accepted units: , m.

7. (2 points) Find the changed patch east-west ground length.

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answer: 1950 m (2 points).

Numerical target 1950; absolute tolerance 3.9. Accepted units: , m.

8. (2 points) Find the changed patch north-south ground length.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 975 m (2 points).

Numerical target 975; absolute tolerance 1.95. Accepted units: , m.

9. (2 points) Find its ground area.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 1901250 m^2 (2 points).

Numerical target 1901250; absolute tolerance 3802.5. Accepted units: , m^2.

10. (2 points) Find its perimeter.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: 5850 m (2 points).

Numerical target 5850; absolute tolerance 11.7. Accepted units: , m.

11. (2 points) Cloud obscures the optical image but the SAR image shows a surface pattern. Explain why these observations differ.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Visible/near-infrared optical observations can be obscured by cloud; radar can often observe through cloud. Interpret radar geometry and surface scattering before classifying the pattern. (2 points).

12. (2 points) A dark SAR patch is labeled water from brightness alone. Give one alternative explanation and a stronger check.

Synthetic Sentinel-2 interpretation exercise: square pixels are 15 m on a side. A rectangular changed patch is 130 pixels east-west and 65 pixels north-south. Map north is up. A paired Sentinel-1 image was acquired through cloud. A separate Landsat image has water, vegetation and built-area classes; all dates, bands and measurements used in each question are stated.

answer: Smooth non-water surfaces, shadow or viewing geometry can also be dark. Compare location, terrain, another date and optical/ground evidence. (2 points).

13. (2 points) A classified map labels 41 of 100 equal-area pixels as forest. Calculate forest cover percentage.

answer: 41 % (2 points).

Numerical target 41; absolute tolerance 0.082. Accepted units: , %.

14. (2 points) Forest pixels decline from 81 to 51. How many equal-area pixels of forest are lost?

answer: 30 (2 points).

Numerical target 30; absolute tolerance 0.06. Accepted units: , .

15. (2 points) Explain why comparing a wet-season image with a dry-season image can mimic land-cover change.

answer: Seasonal vegetation, water and illumination can change reflectance without permanent cover conversion; compare similar seasons and acquisition conditions. (2 points).

16. (2 points) A topographic profile rises 60 m over 1,200 m horizontal distance. Find gradient as a percentage.

answer: 5 % (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , %.

17. (2 points) Two contours are 20 m apart vertically. Adjacent contours are closer horizontally in area A than area B. Which area is steeper and why?

answer: A; the same elevation change occurs over a shorter horizontal distance. (2 points).

18. (2 points) A false-color display makes healthy vegetation red. Does that mean the vegetation reflects red light most strongly? Explain.

answer: No. Display colors are assigned to selected sensor bands; red display can represent near-infrared reflectance. (2 points).

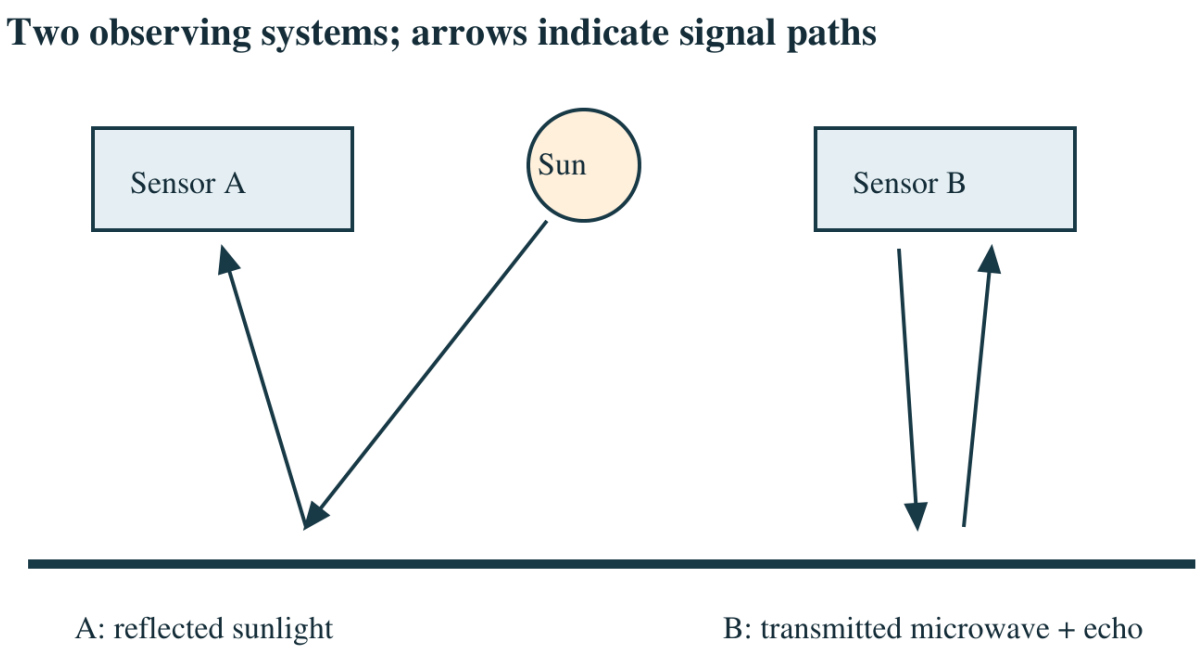
19. (2 points) An ECOSTRESS thermal image shows a warmer paved area than a nearby canopy. Give an observation and a cautious interpretation.

answer: The measured surface-temperature estimate is higher over pavement; material properties, shading and evapotranspiration may contribute. Do not equate surface temperature with air temperature without evidence. (2 points).

20. (2 points) A classifier was trained using labeled examples of water, forest and pavement. Is it supervised or unsupervised? Explain.

answer: Supervised, because labeled training examples associate measurements with classes. (2 points).

21. (2 points) Use Figure F1. Classify A as active or passive using the arrows.



F1. Active and passive observing geometries (original synthetic teaching figure)

answer: Passive: it receives reflected sunlight rather than emitting the probing signal. (2 points).

22. (2 points) Use Figure F1. Classify B as active or passive using the arrows.

Refer to F1. Active and passive observing geometries (original synthetic teaching figure) shown with the first question in this figure group.

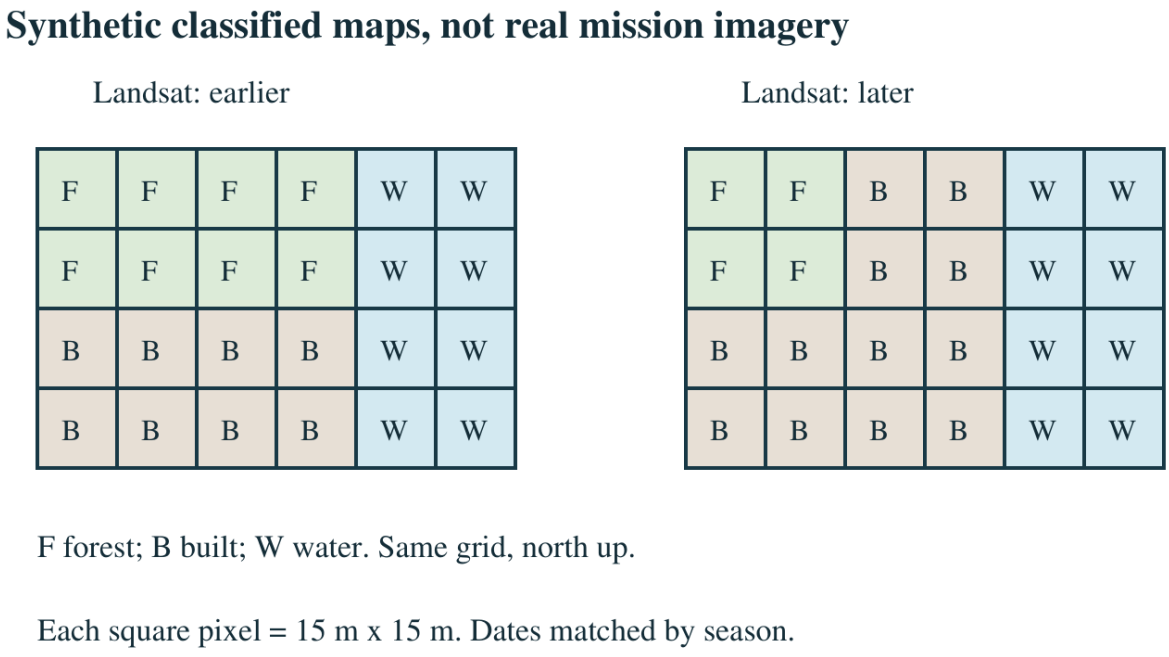
answer: Active: it transmits a microwave signal and measures its return. (2 points).

23. (2 points) Use Figure F1. Which path is generally better suited to observing the surface through cloud, and why?

Refer to F1. Active and passive observing geometries (original synthetic teaching figure) shown with the first question in this figure group.

answer: The microwave radar path B can often penetrate cloud that obscures visible/near-infrared sunlight-based observations; surface scattering and geometry still matter. (2 points).

24. (2 points) Use Figure F2. Which resolution category is quantified by the pixel side length?



F2. Paired classified Landsat maps (original synthetic teaching figure)

answer: Spatial resolution. (2 points).

25. (2 points) Use Figure F2. How many forest pixels change to built class?

Refer to F2. Paired classified Landsat maps (original synthetic teaching figure) shown with the first question in this figure group.

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: .

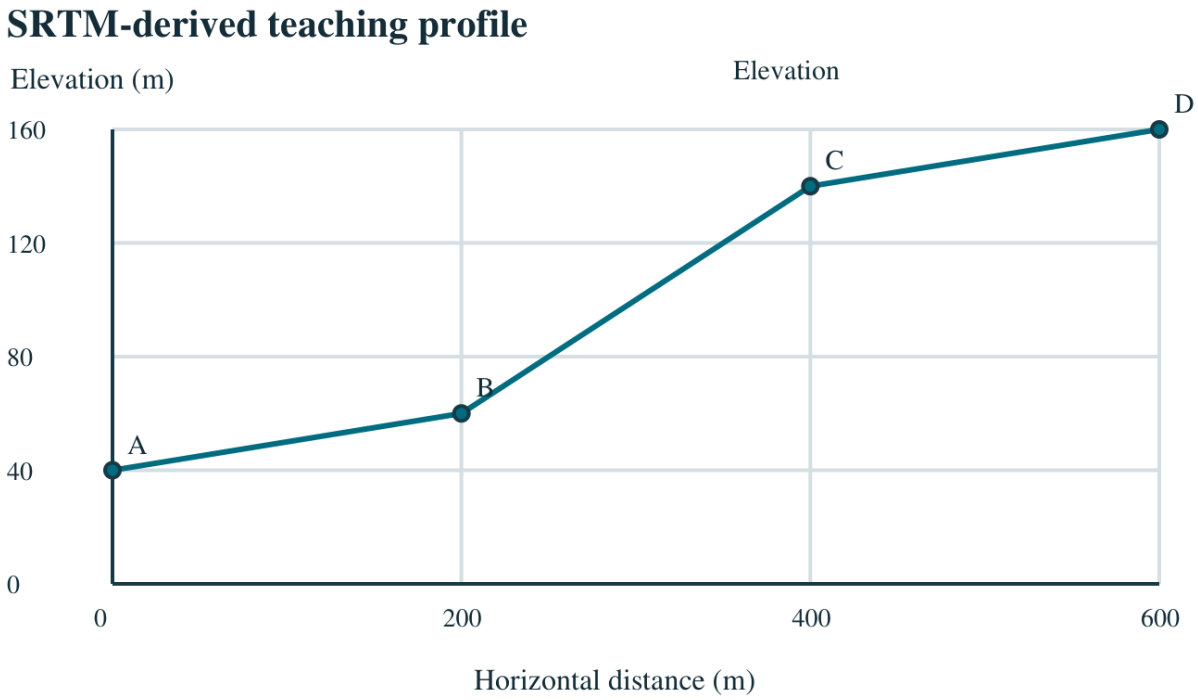
26. (2 points) Use Figure F2. Calculate the ground area of the forest-to-built change.

Refer to F2. Paired classified Landsat maps (original synthetic teaching figure) shown with the first question in this figure group.

answer: 900 m^2 (2 points).

Numerical target 900; absolute tolerance 2.7. Accepted units: , m^2.

27. (2 points) Use Figure F3. Calculate mean slope from B to C as a percentage.



F3. Elevation transect (original synthetic teaching figure)

answer: 40 % (2 points).

Numerical target 40; absolute tolerance 0.12. Accepted units: , %.

28. (2 points) Use Figure F3. Which labeled interval is steepest? Justify from the axes.

Refer to F3. Elevation transect (original synthetic teaching figure) shown with the first question in this figure group.

answer: B to C:80m rise over200m horizontal distance, exceeding the rise per distance in the other intervals. (2 points).

29. (2 points) Use Figure F3. Read the total elevation difference between A and D.

Refer to F3. Elevation transect (original synthetic teaching figure) shown with the first question in this figure group.

answer: 120 m (2 points).

Numerical target 120; absolute tolerance 0.36. Accepted units: , m.

30. (2 points) Use Figure F4. Calculate pavement-minus-canopy surface temperature.

Synthetic ECOSTRESS/Landsat interpretation

Feature	Thermal surface (C)	NIR display channel	Reference class
A	38	bright red	Canopy
B	48	dark red	Pavement
C	25	very dark	Open water

F4. Thermal and false-color products (original synthetic teaching figure)

answer: 10 C (2 points).

Numerical target 10; absolute tolerance 0.03. Accepted units: , C.

31. (2 points) Use Figure F4. Explain why bright red A does not mean its leaves reflect visible red most strongly.

Refer to F4. Thermal and false-color products (original synthetic teaching figure) shown with the first question in this figure group.

answer: The display assigns near-infrared measurements to red; display color is not necessarily the original visible wavelength of reflected light. (2 points).

32. (2 points) Use Figure F4. Can the thermal panel alone establish a 10 C air-temperature difference between A and B? Explain.

Refer to F4. Thermal and false-color products (original synthetic teaching figure) shown with the first question in this figure group.

answer: No. It estimates surface temperature; air temperature requires appropriate atmospheric measurements or a supported surface-air model. (2 points).